Exploratory Data Analysis and Statistical Analysis Report

Diabetes Dataset

Python | Pandas | NumPy | Matplotlib | Seaborn | SciPy

1. Introduction

This project performs Exploratory Data Analysis (EDA) and statistical analysis on a diabetes dataset. The aim is to understand the data structure and quality, explore feature distributions, identify relationships with diabetes, detect duplicates and outliers, and statistically test selected differences between diabetic and non-diabetic groups.

2. Dataset Overview

The original dataset contained 100,000 observations and 9 analytical columns. The target variable is diabetes, where 0 represents no diabetes and 1 represents diabetes.

Item	Result
Original shape	(100,000, 9)
Duplicate rows	3,854
Rows after duplicate removal	96,146
Diabetes = 0	91.5%
Diabetes = 1	8.5%

3. Data Quality Checks

3.1 Missing Values

The dataset was checked for null values. No important missing-value problem was identified, so no imputation was required.

3.2 Duplicate Analysis

Using pandas duplicated(), 3,854 exact duplicate rows were identified. After removing duplicates, 96,146 rows remained. Removing exact duplicates prevents repeated identical observations from receiving extra weight during analysis and later modeling.

3.3 Categorical Features

Gender and smoking_history were treated as categorical features. For correlation calculations, categorical values were numerically encoded. For interpretation, the original smoking-history categories were used: No Info, current, ever, former, never, and not current.

4. Distribution Analysis

Histograms were used to examine numerical feature distributions, particularly age, BMI, HbA1c level, and blood glucose level. They helped identify common value ranges, spread, and unusually high or low observations.

5. Correlation Analysis

A correlation heatmap was used to study linear relationships between features and the diabetes target.

Feature	Correlation with diabetes	Interpretation
Blood glucose level	0.42	Strongest positive relationship
HbA1c level	0.40	Moderate positive relationship
Age	0.26	Weak-to-moderate positive relationship
BMI	0.21	Weak positive relationship
Hypertension	0.20	Weak positive relationship
Heart disease	0.17	Weak positive relationship
Gender	-0.04	Almost negligible

Blood glucose and HbA1c had the strongest positive correlations with diabetes. Correlation indicates association, not causation.

6. Outlier Analysis

Boxplots and the IQR method were used to identify outliers in continuous numerical features.

Feature	Total outliers	Lower	Upper
Age	0	0	0
BMI	7,086	1,121	5,965
HbA1c level	1,315	0	1,315
Blood glucose level	2,038	0	2,038

Upper IQR limits were 38.505 for BMI, 8.3 for HbA1c, and 247.5 for blood glucose. The maximum observed values were 95.69, 9.0, and 300 respectively. These observations were not automatically removed because a statistical outlier is not necessarily an error.

7. Outliers by Diabetes Status

Feature	Upper outliers	Diabetes = 0	Diabetes = 1
BMI	5,965	4,499	1,466
HbA1c	1,315	0	1,315
Blood glucose	2,038	0	2,038

All identified upper outliers for HbA1c and blood glucose belonged to diabetes = 1. For BMI, upper outliers occurred in both groups. This suggests that high HbA1c and glucose values are particularly concentrated in the diabetic class.

8. Blood Glucose vs BMI

The scatter plot showed diabetic observations more concentrated at higher blood glucose levels, while BMI was spread across a wide range in both groups. Blood glucose therefore provided clearer visual separation between diabetes classes than BMI.

9. Average Blood Glucose by Diabetes Status

Group	Mean blood glucose
Non-diabetic	132.85
Diabetic	194.09

The diabetic group had an average blood glucose level about 61.24 units higher than the non-diabetic group. An average is not a diagnostic cutoff; 132.85 should not be treated as a diabetes threshold.

10. Diabetes Percentage by Glucose Range

Glucose range	Diabetes %
0–100	0%
121–140	8.41%
141–160	6.95%
181–200	8.51%
201–220	100%
221–240	100%
241–260	100%
261–300	100%

In this dataset, every observation in the analyzed bins above 200 mg/dL was classified as diabetes = 1. Ranges such as 101–120 and 161–180 were absent because there were no observations in those ranges. This is a dataset-specific pattern, not a general medical cutoff.

11. Target Distribution

Class	Percentage
Diabetes = 0	91.5%
Diabetes = 1	8.5%

The target is highly imbalanced. Therefore, accuracy alone may be misleading during future machine-learning evaluation.

12. Gender vs Diabetes

Gender	No diabetes	Diabetes
0.0	90.25%	9.75%
1.0	92.38%	7.62%

The two gender categories had relatively similar diabetes percentages, consistent with the almost negligible correlation of -0.04.

13. Smoking History vs Diabetes

Smoking history	No diabetes	Diabetes
No Info	95.94%	4.06%
Current	89.79%	10.21%
Ever	88.21%	11.79%
Former	83.00%	17.00%
Never	90.47%	9.53%
Not current	89.30%	10.70%

Former smokers had the highest observed diabetes percentage at 17.00%, while No Info had the lowest at 4.06%. This shows an association in this dataset and does not prove causation.

14. Statistical Analysis

Two independent-samples t-tests were performed using Welch's t-test (equal_var=False) to compare diabetic and non-diabetic groups.

14.1 Hypothesis Test: Blood Glucose vs Diabetes

H0: There is no significant difference in mean blood glucose between diabetic and non-diabetic groups.
H1: There is a significant difference in mean blood glucose between the groups.

Statistic	Result
Non-diabetic mean	132.85
Diabetic mean	194.09
T-statistic	-94.7949
P-value	0.0 displayed; extremely small
Decision	Reject H0

Because the p-value is far below 0.05, the null hypothesis is rejected. There is a statistically significant difference in mean blood glucose between the two groups.

14.2 T-test: BMI by Diabetes Status

H0: There is no significant difference in mean BMI between diabetic and non-diabetic groups. H1: There is a significant difference in mean BMI between the groups.

Statistic	Result
Non-diabetic mean BMI	26.89
Diabetic mean BMI	31.99
T-statistic	-60.2651
P-value	0.0 displayed; extremely small
Decision	Reject H0

Because the p-value is far below 0.05, the null hypothesis is rejected. There is a statistically significant difference in mean BMI between the two groups. The diabetic group had a mean BMI about 5.10 units higher.

15. Key Insights

- Blood glucose had the strongest correlation with diabetes ($r = 0.42$).
- HbA1c had the second strongest correlation ($r = 0.40$).
- Mean glucose was 132.85 in non-diabetic observations and 194.09 in diabetic observations.
- The glucose t-test showed a statistically significant group difference ($t = -94.79$, $p < 0.001$).
- Mean BMI was 26.89 in non-diabetic observations and 31.99 in diabetic observations.
- The BMI t-test showed a statistically significant group difference ($t = -60.27$, $p < 0.001$).
- The target was imbalanced: 91.5% non-diabetic and 8.5% diabetic.
- All observations in the analyzed glucose bins above 200 mg/dL were diabetes = 1.
- Former smokers had the highest observed diabetes percentage among smoking-history categories (17%).

16. Overall Conclusion

The EDA and statistical analysis provided a clear picture of the dataset. Blood glucose and HbA1c were the most prominent features in the correlation analysis. BMI also showed a statistically significant difference between diabetic and non-diabetic groups. Gender showed very little linear association, while smoking-history categories showed different diabetes percentages.

The dataset is imbalanced, so future machine-learning models should be evaluated with metrics such as precision, recall, F1-score, and a confusion matrix rather than accuracy alone. The next stage can include final categorical encoding, feature-target separation, train-test splitting, scaling where appropriate, model training, and evaluation.

17. Interpretation Notes and Limitations

- Correlation does not establish causation.
- A statistically significant difference does not automatically mean the difference is clinically important.
- The glucose-above-200 pattern is specific to this dataset and should not be presented as a universal diagnostic rule.
- The software displayed p-values as 0.0 because they were extremely small; reporting $p < 0.001$ is more appropriate.
- Outliers were identified statistically and were not automatically considered errors.
- Numerically encoded categorical values should not be interpreted as naturally ordered categories.